

Leukemia Subtyping

Sorting Blood-Smear Cells — and Why 99.8% Is a Warning

Capstone Project Report

A vision model that subtypes leukemic cells from blood smears — and a case study in why a near-perfect score can be the least trustworthy result.

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1. Executive Summary

This project sorts a single white blood cell from a smear into four classes — Benign plus three stages of B-cell acute lymphoblastic leukaemia (Early Pre-B, Pre-B, Pro-B) — with a transfer-learned vision model. It scores **99.8% accuracy** on held-out images.

That 99.8% is presented as a **red flag, not a triumph**. The dataset ships no patient identifiers, so a random split almost certainly puts the same patient's cells in both training and test — and the model can win on staining and patient artefacts rather than on leukaemia biology.

2. Introduction & Motivation

Distinguishing benign look-alike cells (hematogones) from leukemic blasts, and telling the ALL subtypes apart, is subtle work even for specialists — a genuinely interesting vision problem. It is also the most clinically serious task in the series, so the honest evaluation matters most here.

3. Data

The dataset (Aria et al., 2021, via Kaggle) contains 3,256 peripheral-blood-smear images of one cell each, across the four classes. Images were split 70/15/15 into train/validation/test, stratified by class.

The unavoidable caveat

This public release contains **no patient IDs** (filenames are just sequential), so a strict patient-level split is impossible. Some patient leakage between splits is therefore likely, and pretending otherwise would be the dishonest move.

4. Methodology

The classifier is a YOLO11s-cls network (5.4M parameters), transfer-learned from ImageNet — the same computer-vision pipeline as MoleCheck, reused directly. Training reached 100% validation accuracy within ~30 epochs.

5. Results

Metric	Value
Test accuracy	99.8%
Macro-F1	0.998
Errors	1 of 489 images

On most projects this would be the headline. Here it is the cue to look harder at the split: cells from one slide share staining, lighting, focus and background, and a strong model can score near-perfectly by recognising **those** cues.

6. Deployment

Exported to ONNX (21 MB) for an in-browser demo (pick a real cell image, classify it on-device) and to TensorFlow Lite for an Android app ('Cell Explorer'), with predictions verified to agree with the reference model.

7. Limitations & Ethical Considerations

- Not a diagnostic tool — leukaemia diagnosis needs bone-marrow analysis, flow cytometry and genetics.
- The headline accuracy is inflated by likely patient leakage; treat it as an upper bound.
- A single, specific dataset; real clinical variety is far greater.

8. Future Work

The clear next step (v2) is the **C-NMC 2019** dataset, which ships patient IDs for ~15,000 cells. Holding out whole patients — as done in the EEG project — will give a lower but far more trustworthy number. That honest result is worth more than this one.